When Behavioral Traces Mislead: Evaluating Evidence Adequacy in Learning Analytics

Keywords: behavior profiling, learning analytics, learning management system, evidentiary adequacy, governance

Extended Abstract

The transition from high school to college requires students to shift from externally guided behaviors to independent self-regulated learning [1]. In large-enrollment computing courses, learning management system data provides scalable traces of how students distribute coursework across a semester. Such temporal behaviors are widely used in learning analytics systems to infer risk and trigger interventions [2], [3], [4].

This practice reflects a broader assumption: observable behavioral traces can stand in for latent constructs such as motivation, capacity, or self-regulation. However, the **evidentiary adequacy** of these temporal signals remains unclear [5], [6], [7]. Students who submit close to deadlines are frequently flagged as procrastinating or at risk to trigger automated nudges or instructor attention [2], [3], [4], [8], [9]. In fact, learners may actually adopt diverse strategies reflecting differences in work schedules, prior knowledge, and external constraints [10], [11]. **Interpreting timing behavior primarily as a single latent construct of procrastination often oversimplifies the reality of student self-regulation.**

To address this gap and shift from examining timing as a simple proxy to understanding repeated time-based self-regulation patterns, this study asks: *Do observable temporal submission patterns meaningfully distinguish learners in ways that justify individual-level interventions?*

We analyze 32,275 assignment submission records from 4,027 enrollments (3,982 unique students) from across 4 introductory computing courses over 6 semesters.

Contribution 1: We identified four distinct *learner profiles* based on the Submission Fraction Index (SFI) – Consistent Early, Gradual Early, Gradual Late, Consistent Late – and examined their relationship with group academic outcomes. Due to the nature of varied assignment types, submission timestamps are not directly comparable. We therefore introduce the Submission Fraction Index (SFI), a normalized measure capturing the proportion of the available assignment window that has elapsed at the time of submission. Normalizing submission time in this way enables meaningful comparison across heterogeneous assignments, capturing students' temporal regulation behavior relative to the unique structure of each assignment.The learner profile cluster centroids across resolutions (k = 3–5) are shown in Figure 1, with silhouette scores ($s \approx 0.27 - 0.28$) indicating moderate but consistent structure (Table 1). Projection into slope-mean space (Figure 2) reveals separable behavioral regions, and subsampling analyses (Figure 4) reproduce the same profiles across enrollment-level and course-level splits. While earlier and more consistent submission behavior is associated with moderately higher academic performance at the group level, the grade outcome distributions substantially overlap across all profiles. Engagement measures such as activity time, page views, and participations show negligible separation (Figure 3, Table 1). ▷ **Finding 1**: Temporal submission behavior exhibits reproducible differentiation at scale; however, timing behavior alone is insufficient to support intervention decisions due to high outcome overlap.

Contribution 2: We evaluated whether temporal pacing trajectories provide sufficient signal for reliable individual-level academic prediction. Empirical results shown in Table 2 demonstrate the limited predictive power of pacing metrics, which becomes critical when making individual-level performance predictions. We trained supervised learning models—including Decision Tree, Random Forest, XGBoost, and a 1D Convolutional Neural Network—to predict individuals' final course score outcomes using their SFI trajectories. Predictive models trained solely on submission timing explained very little individual performance variance ($R^2 \approx 0.05$) across cross-validation. ▷ **Finding 2**: The SFI-based learner profiles demonstrate limited capacity for predicting individual academic performance.

Contribution 3: We release open-source artifacts to support reproducibility & research, and propose a set of design implications. We contribute open-source artifacts, at https://anonymous.4open.science/r/SRL-Learner-Profiles-DD47/, including data preprocessing pipelines, SFI computation utilities, clustering implementations, and modeling scripts to support replication and future research of temporal behavioral analyses in computing education. Our findings point to a misalignment between the resolution of available measurements and the interventions those measurements are used to justify. We propose the following design implications:

1. *Design for reflective feedback, not automated categorization.* Temporal pacing patterns can inform lightweight, student-facing supports such as deadline planning prompts, pacing visualizations, or optional reminders. However, given substantial overlap in performance distributions across profiles, these patterns should not trigger high-stakes categorization or automated risk labeling. Submission trajectories are most appropriate as descriptive tools that scaffold self-regulation – as suggested by [12] – rather than as prescriptive indicators of academic standing.

2. *Constrain high-stakes interventions based solely on behavioral trace data.* Behavioral traces alone are insufficient for high-stakes automated interventions as they explain only a small portion of performance variance. This aligns with prior work suggesting that academic performance may be governed by unobserved structural constraints rather than observable behavioral regularities alone [13]. Systems that act on timing data alone risk misclassifying students whose pacing behavior reflects external constraints rather than regulatory difficulty. Therefore, not only a methodological concern but an ethical safeguard against algorithmic inference.

3. *Architect privacy-aware personalization through student-controlled contextual disclosure.* Intervention systems should provide mechanisms for students to contextualize their pacing behavior through voluntary disclosure of relevant constraints (e.g., employment, caregiving, course load). Rather than exposing such information to instructors by default, personalization logic can operate within student-defined privacy boundaries, using an approach like that of [14]. Under this model, students retain agency over what contextual information is shared, while still benefiting from personalized learning—like [15], [16]—systems tailored to their circumstances.

Temporal behavioral traces provides a pathway to separate students into patterns. But separating patterns is not the same as justifying interventions. Our results show that clear differences in temporal pacing do not necessarily translate into clear differences in academic performance outcomes. As AI systems increasingly automate interpretation and personalized intervention, weak signals can quickly become fixed labels. Knowing what data cannot support is as important as knowing what it can.

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Table 1: The numerical results of clustering analyses for 3, 4, and 5 clusters: cluster IDs are ordered by the average SFI in ascending order. s indicates the silhouette score.

		3 clusters ($s = 0.2785$)			4 clusters ($s = 0.2734$)				5 clusters ($s = 0.2798$)				
Metric	Measure	cluster 0	cluster 1	cluster 2	cluster 0	cluster 1	cluster 2	cluster 3	cluster 0	cluster 1	cluster 2	cluster 3	cluster 4
Enrollments	count	1259	1295	1473	1110	1250	617	1050	450	910	1046	587	1034
Standardized course grade	median	0.38	0.15	-0.03	0.40	0.15	0.21	-0.13	0.45	0.32	0.13	0.21	-0.14
	IQR	1.15	1.24	1.37	1.13	1.24	1.23	1.43	1.01	1.19	1.26	1.22	1.42
	Cliff's delta with cluster 0	0	-0.16	-0.26	0	-0.17	-0.13	-0.32	0	-0.11	-0.24	-0.19	-0.38
Standardized activity time	median	-0.23	-0.22	-0.25	-0.23	-0.22	-0.16	-0.30	-0.24	-0.23	-0.21	-0.16	-0.30
	IQR	0.98	1.00	1.01	0.99	1.00	1.14	0.97	0.92	1.06	0.99	1.12	0.97
	Cliff's delta with cluster 0	0	0.02	-0.02	0	0.02	0.05	-0.06	0	-0.02	0.02	0.03	-0.07
Standardized page views	median	-0.17	-0.18	-0.23	-0.17	-0.18	-0.13	-0.29	-0.09	-0.18	-0.19	-0.12	-0.29
	IQR	1.02	1.08	1.10	0.99	1.07	1.20	1.09	1.09	0.98	1.08	1.21	1.09
	Cliff's delta with cluster 0	0	-0.01	-0.05	0	-0.02	0	-0.09	0	-0.06	-0.06	-0.04	-0.13
Standardized participations	median	-0.04	-0.08	-0.09	-0.04	-0.08	-0.03	-0.13	0.00	-0.07	-0.07	-0.03	-0.13
	IQR	1.06	1.01	1.13	1.07	1.02	1.14	1.09	1.08	1.06	1.02	1.15	1.09
	Cliff's delta with cluster 0	0	-0.02	-0.07	0	-0.03	-0.01	-0.08	0	-0.06	-0.05	-0.05	-0.12

Table 2: The 3-fold cross-validation results of four supervised models. Each model was validated on two different folding-strategies, and each on 2–3 different interpolation strategies. Bold numbers indicate the best result among models and interpolation strategies.

		by enrollments		by courses	
Model	Interpolation	RMSE	R^2	RMSE	R^2
Decision Tree	linear	0.9756	0.0213	0.9752	0.0229
	zero-fill	0.9853	0.0008	0.9972	-0.0204
Random Forest	linear	0.9593	0.0542	0.9570	**0.0578**
	zero-fill	0.9661	0.0406	0.9666	0.0377
XGBoost	linear	0.9563	**0.0600**	0.9599	0.0523
	zero-fill	0.9701	0.0309	0.9664	0.0418
	NaN-fill	0.9625	0.0469	0.9638	0.0456
1D-CNN	linear	0.9648	0.0421	0.9658	0.0403
	zero-fill	**0.9476**	0.0238	**0.9527**	0.0131

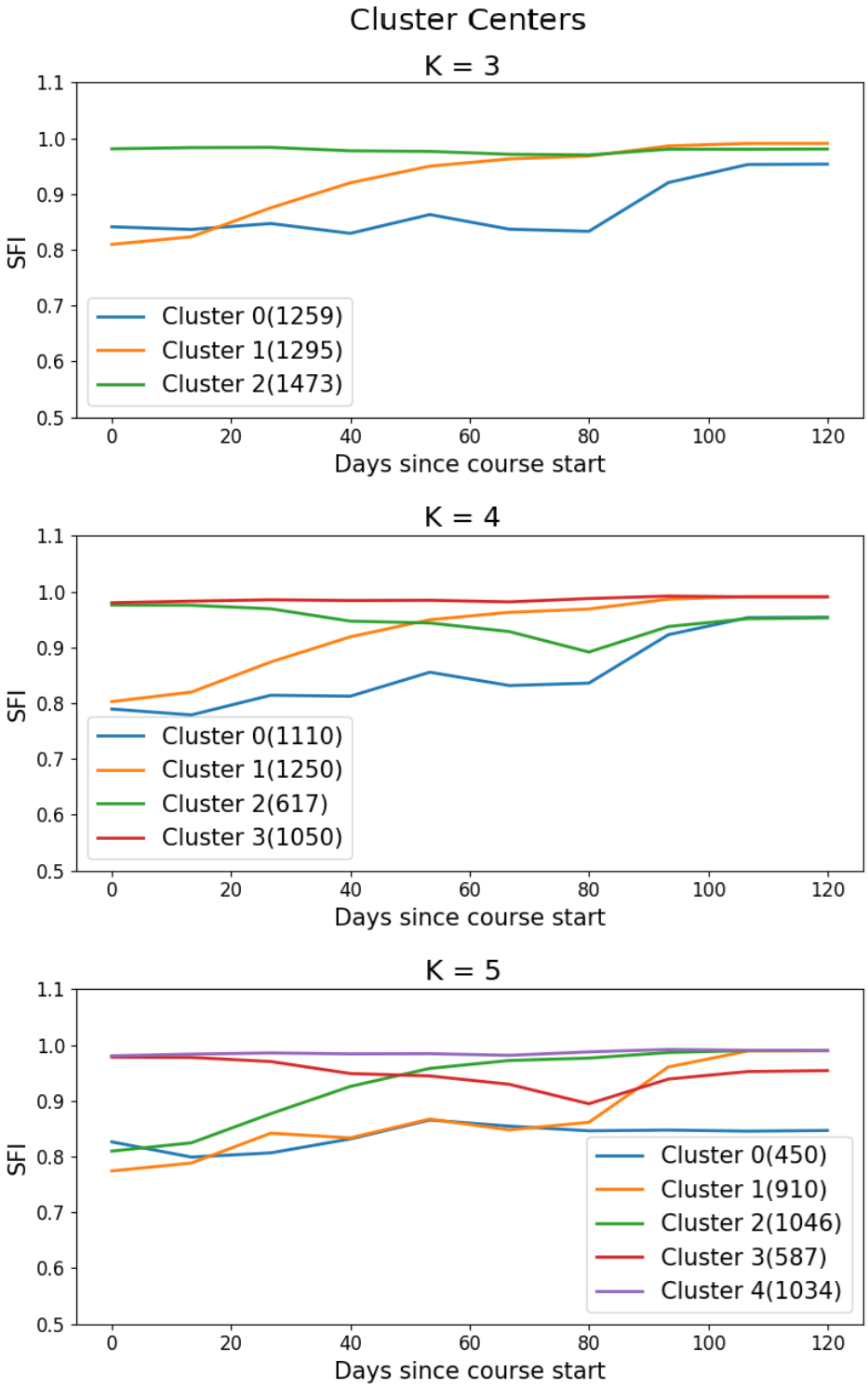

Figure 1: Cluster centroids for three clustering resolution, plotted in time-SFI dimension. The numbers in each parenthesis indicates the number of enrollments assigned to the cluster.

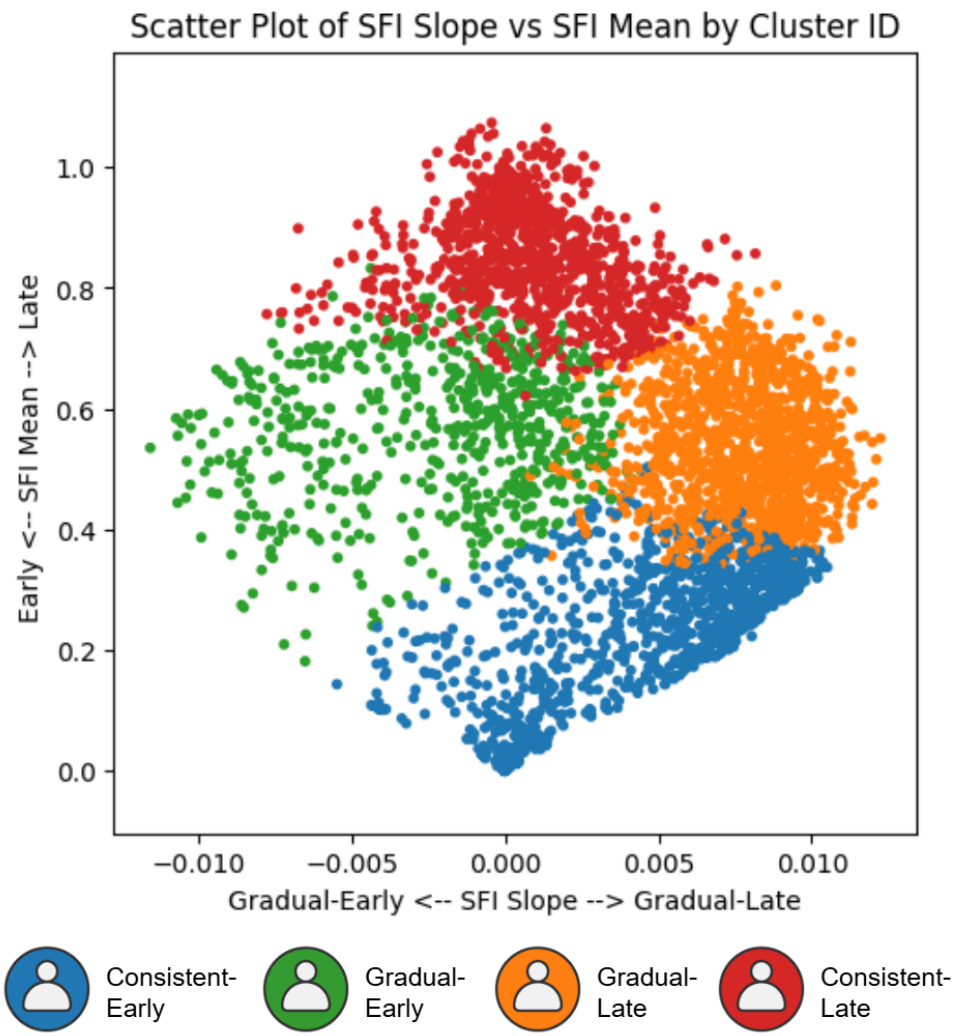

Figure 2: Enrollment-level submission trajectories projected onto mean–slope space. The x-axis represents the linear slope of SFI across the semester (temporal trend), and the y-axis represents average SFI (overall submission timing). Each point represents an enrollment.

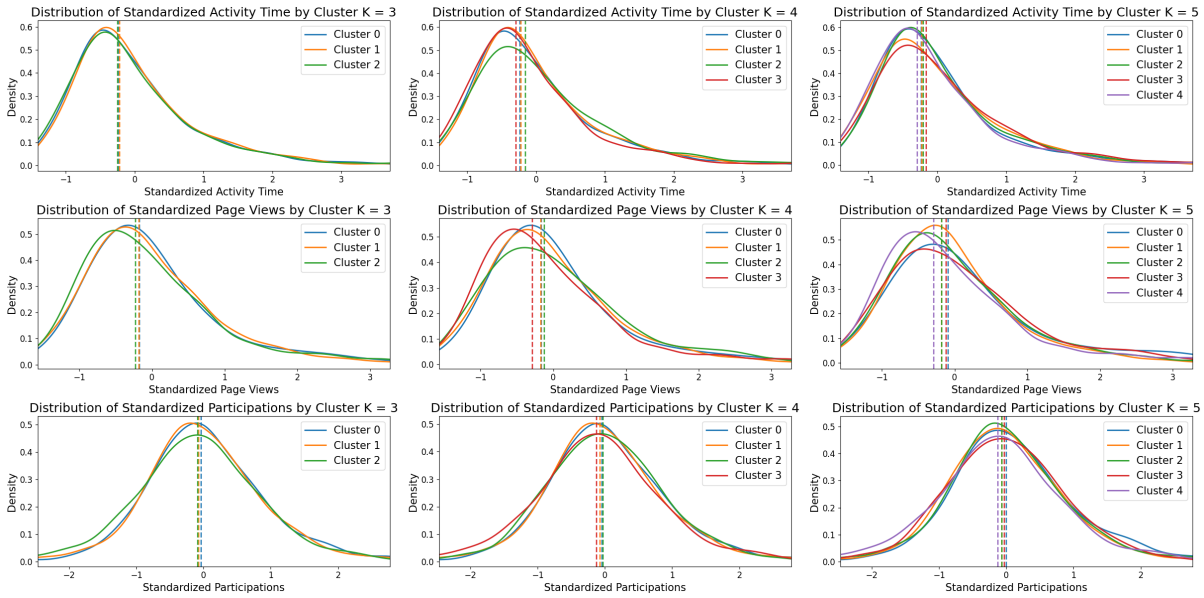

Figure 3: Probability density distribution and the median of three behavior metrics for each cluster resolution. Plots for standardized activity time, page views, and participations are presented in each rows, top to bottom.

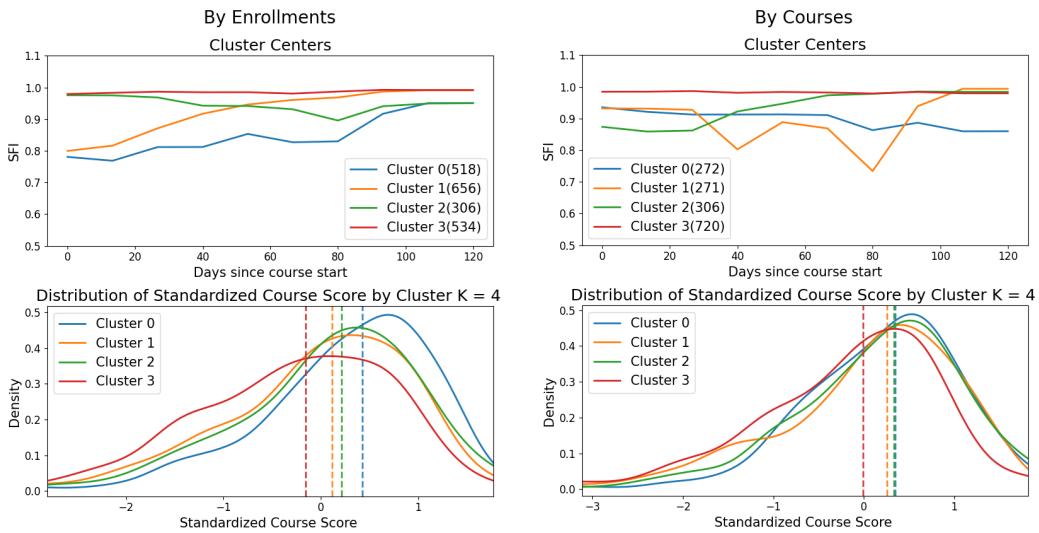

Figure 4: Four-cluster solutions for two 50% subsampled datasets. For each subsample, cluster centroids (SFI trajectories over time) are shown alongside probability density distributions of standardized course grades. Results are shown for enrollment-level subsampling (left panels) and course-level subsampling (right panels). The similar four temporal profiles and grade-ordering patterns persist across subsamples, demonstrating robustness.